An Explainable Multi-Stage Decision-Support Pipeline for Chest X-Ray Classification, Localization, and Structured Reporting

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Abstract—Chest X-ray analysis is clinically useful only when an automated system provides more than an image-level abnormality score. A deployable decision-support workflow should couple classification, spatial evidence, and a concise explanation that a clinician can audit. This work presents a multi-stage pipeline for a 15-label subset of the VinBigData chest X-ray dataset. The pipeline combines image-level multi-label classification, YOLO-based localization, and an agentic reporting layer that converts model outputs into a structured decision-support summary. The study compares a lightweight CNN trained from scratch with ImageNet-pretrained EfficientNet-B3, ResNet-50, Vision Transformer, and Swin Transformer models. Swin achieves the strongest image-level performance with a macro AUC-ROC of 0.9665, while a 1.31M-parameter CNN remains competitive at 0.9314. The best retained YOLOv8m localization run reaches 0.3666 mAP@0.5. On a 300-case review slice, the agentic reporting workflow attains 0.750 exact-match accuracy and 0.9560 macro accuracy, while producing standardized summaries grounded in classifier probabilities and detector outputs. The resulting system is best interpreted as an explainable decision-support pipeline rather than a standalone diagnostic model.

Index Terms—Chest X-ray, VinBigData, multi-label classification, Swin Transformer, YOLOv8, explainable AI, clinical decision support, structured reporting.

I. Introduction

Chest radiography remains one of the most common imaging studies in routine clinical care. Consequently, machine learning systems for chest X-ray analysis are most useful when they support a clinician-facing workflow rather than only maximize a leaderboard metric. In practice, a clinically readable system should answer at least three questions: what findings are predicted, where the relevant image evidence is located, and how the evidence should be summarized for review.

This project was therefore framed as a decision-support problem rather than as a pure classification benchmark. The resulting workflow combines three components: (1) multi-label image classification, (2) bounding-box localization, and (3) a structured reporting layer that organizes the evidence into a standardized summary. The design goal is not to replace radiologist interpretation, but to provide a more interpretable intermediate product than raw probabilities alone.

The work makes three practical contributions:

- a reproducible comparison between a from-scratch CNN baseline and several transfer-learning classifiers on a 15-label VinBigData subset,
- a YOLOv8m localization branch trained on the available VinBigData bounding-box annotations, and
- an agentic reporting layer that merges classifier probabilities and detector outputs into a concise decision-support summary.

II. Related Work

Chest X-ray AI has already been studied extensively across classification, localization, and report generation. In image-level classification, CheXNet demonstrated the practical value of deep convolutional transfer learning for chest radiograph interpretation [2]. On the VinDr side, Pham et al. introduced an explainable VinDr-CXR system that explicitly combined multi-label classification with lesion localization and showed that such assistance could improve interobserver agreement among readers [3].

Automated chest X-ray report generation has also been studied before the recent wave of multimodal LLMs. Fine-grained label learning [9], contrastive attention [10], and variational topic inference [11] all aim to improve how visual findings are translated into clinically plausible report text. More recent multimodal approaches such as CXR-LLaVA [12] and reader studies on GPT-4 impression generation [13] show that large vision-language models can now produce readable chest X-ray summaries, but they are usually trained and evaluated on large report corpora such as MIMIC-CXR rather than on a label-and-box dataset like the retained VinBigData subset used here.

Our project therefore does not claim to be the first to combine imaging models with language generation. Instead, the distinguishing choice is pragmatic: rather than train a new end-to-end report generator, we start from a bounded VinBigData subset with image-level labels and localization annotations, then assemble a reproducible decision-support workflow that keeps the classifier, detector, and reporting stages explicitly separable and auditable.

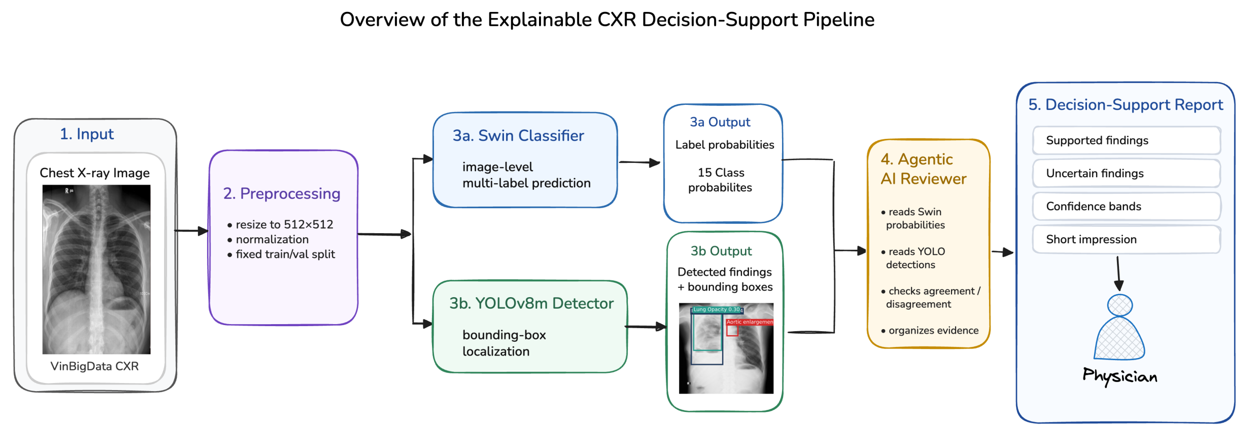

Fig. 1. Locked workflow used in the final repository: chest X-ray image, Swin classification, YOLO localization support, and agentic AI report generation.

III. Dataset and Task Formulation

The study uses the current 15-label VinBigData subset available in the repository. The retained labels are Aortic enlargement, Atelectasis, Calcification, Cardiomegaly, Consolidation, ILD, Infiltration, Lung Opacity, Nodule/Mass, Other lesion, Pleural effusion, Pleural thickening, Pneumothorax, Pulmonary fibrosis, and No finding. The image-level task is multi-label classification because a single radiograph may contain multiple concurrent findings.

For localization, the raw VinBigData bounding-box annotations are used to train a YOLOv8m detector. The classification and localization tasks are connected only downstream in the reporting workflow. They are evaluated separately because they answer different clinical questions: classification predicts whether a finding is present anywhere in the image, while localization tests whether the model can identify spatially plausible supporting regions.

The repository uses a fixed train-validation split of 12,000 training images and 3,000 validation images. This split is reused across the retained experiments to support comparability and reproducibility.

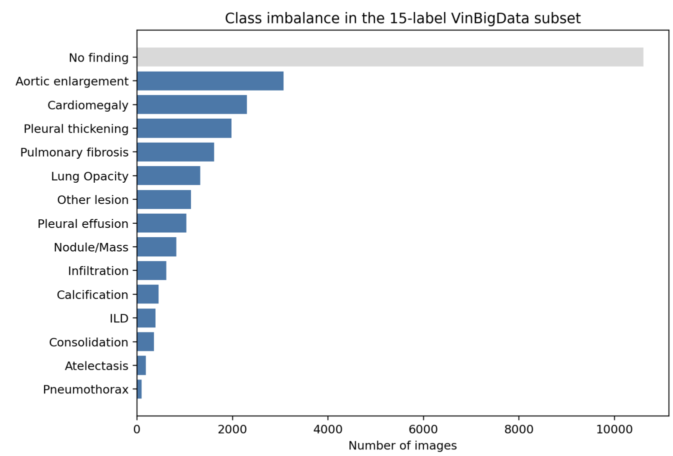

Fig. 2. Label frequency in the retained 15-label VinBigData subset. The dataset is strongly imbalanced, with No finding dominating and several abnormalities occurring rarely.

IV. Methods

A. Image-Level Classification

All classification models were trained under a unified multi-label setup. Input chest X-rays were resized to 512×512, the data split was fixed, and training used binary cross-entropy with logits:

$$\mathcal{L}_{\mathrm{BCE}} = -\frac{1}{NC}\sum_{i=1}^{N}\sum_{c=1}^{C}\Big[y_{ic}\log\sigma(z_{ic}) + (1-y_{ic})\log(1-\sigma(z_{ic}))\Big], \quad (1)$$

where N is the batch size, $C = 15$ is the number of labels, y_{ic} is the binary ground-truth label, and z_{ic} is the model logit for class c.

The retained classifiers are:

- simple_cnn: a 1.31M-parameter CNN trained from scratch,
- efficientnet_b3: an ImageNet-pretrained EfficientNet-B3 with a task-specific head,
- resnet50: an ImageNet-pretrained ResNet-50 with a task-specific head,
- vit_base: a Vision Transformer baseline, and
- swin_base: a hierarchical Swin Transformer.

This benchmark isolates two practical questions: whether strong performance requires a large transformer, and how much the pipeline benefits from transfer learning relative to a lightweight from-scratch baseline.

B. Localization with YOLOv8m

Localization uses YOLOv8m trained on the retained VinBigData bounding boxes. YOLO is evaluated as a detector using mAP@0.5, and its detections are also converted into image-level labels so that the detector can be compared to the image-level classifier as a supporting source of evidence. The latter evaluation is not intended to replace the detection metric; it quantifies how much clinically relevant label signal is recoverable from bounding boxes alone.

C. Agentic Reporting Layer

The final stage is an agentic review layer:

$$\begin{aligned} \text{CXR} &\rightarrow \text{classifier} \rightarrow \text{detector} \\ &\rightarrow \text{agentic AI} \rightarrow \text{structured report.} \end{aligned} \tag{2}$$

The reporting model does not operate directly as a diagnostic classifier. Instead, it consumes (1) the full classifier probability vector, (2) thresholded predicted labels, (3) YOLO detections and confidences, (4) bounding boxes, and (5) classifier-detector agreement or disagreement. The resulting report contains supported findings, uncertain findings, a short impression, review guidance, and a safety statement.

To improve readability, the report also maps low-level findings into a small set of derived global buckets, including No acute abnormality, Cardiomediastinal abnormality, Pleural abnormality, and Chronic interstitial or fibrotic pattern. These are interpretive summaries only; they are not treated as additional supervised labels.

V. Experimental Setup

The retained training scripts use fixed defaults to support reproducibility. The from-scratch CNN was trained for 100 epochs. The transfer-learning classification runs were executed for 10 epochs each. The retained YOLO experiments include an initial 10-epoch run and longer continuation runs, with the best retained detector checkpoint reported in the results. The classification models were trained with AdamW, a learning rate of 10^{-4}, weight decay of 10^{-4}, and cosine scheduling. The primary classification metric is macro AUC-ROC.

All experiments were run on the oc4 compute environment. The final reproduction workflow is structured so that classification and localization runs can be launched consistently from the cleaned repository layout.

VI. Results

A. Image-Level Classification

Table I and Fig. 3 summarize the image-level classification benchmark. Swin is the strongest image-level classifier with a macro AUC-ROC of 0.9665. EfficientNet-B3 and ResNet-50 are also strong, indicating that transfer learning from natural-image pretraining remains effective in this setting. The most notable baseline result is the from-scratch CNN, which achieves 0.9314 macro AUC-ROC with only 1.31M parameters. By contrast, the ViT baseline underperforms in this setup.

Two observations follow. First, strong performance does not require a transformer by default: the CNN baseline is substantially better than a weakly configured ViT path and is not far from the transfer-learning baselines.

TABLE I
Image-Level Classification Results on the 15-Label VinBigData Subset

Model	Params	Epochs	Macro AUC-ROC
simple_cnn	1.31M	100	0.9314
efficientnet_b3	11.49M	10	0.9564
resnet50	24.56M	10	0.9550
vit_base	86.84M	10	0.8143
swin_base	87.28M	10	0.9665

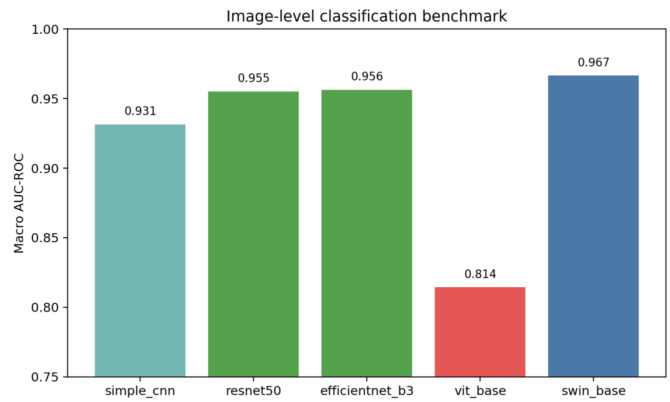

Fig. 3. Minimal comparison of the retained image-level classifiers. Transfer learning helps substantially, but the lightweight from-scratch CNN remains a strong sanity-check baseline.

Second, architecture still matters at the high end: Swin's hierarchical design appears better matched to this problem than the ViT configuration used here.

B. Localization Performance

YOLO is intentionally treated as a supporting localization model rather than a competing image-level classifier. The best retained YOLOv8m run reached 0.3666 mAP@0.5, while a larger YOLOv8l variant did not materially improve on that result. This reinforces the final design decision: Swin remains the primary visual model, and YOLO is used to provide spatial evidence that can confirm, qualify, or challenge a classifier prediction inside the reporting stage.

C. End-to-End Agentic Workflow

The agentic reporting workflow was evaluated on a 300-case review slice. Table II shows that the reporting pipeline slightly improves exact-match accuracy and macro accuracy relative to the individual components, while remaining below Swin on macro F1. This is consistent with the intended role of the layer: it is primarily an explainability and communication module, not a new visual backbone.

The same 300-case study also provides a more interpretable operational view. The agentic layer improved over Swin on 16 cases and over YOLO on 33 cases, while underperforming Swin on 12 cases and YOLO on 9 cases. These numbers suggest modest quantitative benefit at the end-to-end level, but the stronger value of the

TABLE II
300-Case Comparison of the End-to-End Decision-Support Workflow

System	Exact-Match	Macro Acc	Macro F1
Swin classifier	0.7400	0.9542	0.5671
YOLO-derived labels	0.7300	0.9464	0.3286
Agentic AI workflow	0.7500	0.9560	0.4855

reporting stage is qualitative: it organizes raw predictions into a clinician-readable summary with explicit support and uncertainty.

D. Qualitative Case Studies

Representative outcomes from the retained report set illustrate where the workflow is most useful in practice.
Normal case (‘302b0d07…‘). Ground truth, Swin, YOLO, and the final report all indicated No finding. This is the simplest workflow setting: the system produces a conservative normal summary without forcing unnecessary explanation.

Clean cardiomediastinal case (‘a3d5137d…‘). The radiologist labels were Aortic enlargement and Cardiomegaly. Both Swin and YOLO agreed exactly, making the final report straightforward to support with both image-level and localized evidence.

Multi-finding case (‘127f6676…‘). The radiologist labels were Aortic enlargement, Cardiomegaly, Lung Opacity, and Nodule/Mass. Swin recovered all four findings, while YOLO localized the three more focal patterns and missed the more diffuse Lung Opacity. This is a representative example of the intended division of labor: the classifier captures the broader abnormal picture, while the detector contributes localization support where spatial evidence is most meaningful.

VII. Discussion

The central result is that the project succeeds more as a workflow than as a claim of architectural novelty. Swin provides the strongest image-level performance, YOLO adds a useful but imperfect localization signal, and the agentic review layer makes the combined output easier to interpret. This is meaningful in a pragmatic clinical setting because a physician benefits less from an isolated probability vector than from a compact summary that states likely findings, shows where the local evidence is, and makes uncertainty explicit.

The CNN baseline is also important to the scientific story. A 1.31M-parameter model trained from scratch reached 0.9314 macro AUC-ROC, which prevents the work from overstating the necessity of large transformers for this dataset. Conversely, the Swin result shows that stronger pretraining and architectural bias still matter at the top end of the benchmark.

The localization branch provides complementary value rather than replacing classification. Its detection mAP remains modest, but the branch becomes more useful when framed as supporting evidence within a structured report. This is particularly true for localized or focal abnormalities. The detector is less reliable for diffuse abnormalities, which is consistent with both the metric behavior and the qualitative case studies.

From a workflow perspective, this means the system is best viewed as a first-pass assistant. A clinician can use the classifier output to understand the likely abnormal pattern, use the detector overlay to inspect whether there is localized support, and then use the agentic summary as a structured starting point for review rather than a final autonomous report.

VIII. Limitations

Several limitations should be made explicit. First, this study uses a retained 15-label VinBigData subset rather than the full original taxonomy. Second, the agentic layer has not been clinically validated against radiologist-authored free-text reports. Third, the report-generation stage remains dependent on upstream model quality; it does not correct all classification or localization errors. Fourth, the current analysis emphasizes aggregate metrics and does not yet include calibration studies, external validation, reader studies, or subgroup analyses.

These limitations are important for publication framing. The present system should be interpreted as an explainable decision-support prototype with a reproducible code path, not as a clinically deployable autonomous reporting system.

IX. Conclusion

This work presents a practical multi-stage chest X-ray decision-support pipeline built on the VinBigData dataset subset available in the repository. Swin is the strongest image-level classifier, a lightweight CNN provides a strong sanity-check baseline, YOLO contributes localization evidence, and the agentic reporting layer converts model outputs into a structured summary that is easier to audit than raw probabilities alone. The resulting system is therefore best described as a clinically motivated, explainable workflow for chest X-ray analysis rather than as a new standalone diagnostic model.

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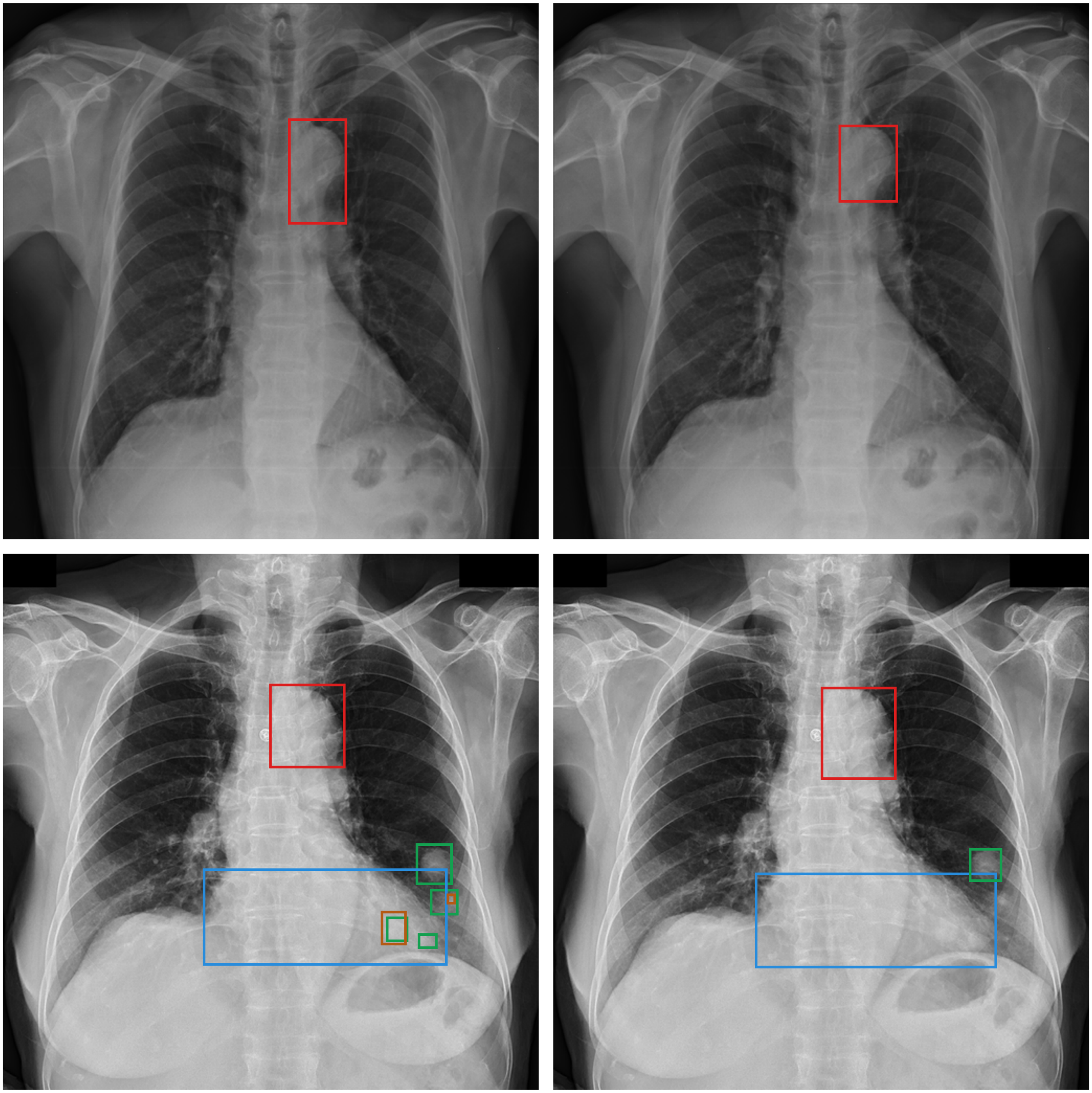

Fig. 4. Ground-truth-versus-YOLO comparison examples. In each row, the left panel shows merged radiologist ground-truth boxes and the right panel shows YOLO predictions. Top row: a single-label case where both the ground truth and YOLO indicate Aortic enlargement. Bottom row: a multi-label case with radiologist labels Aortic enlargement, Cardiomegaly, Lung Opacity, and Nodule/Mass; YOLO recovers the first three localizable patterns except the more diffuse Lung Opacity finding.